Shi Luode

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Research Interests: Embodied AI and Robot Learning, Reinforcement Learning for Robotics, and Neural Combinatorial Optimization

EDUCATION

National University of Singapore (NUS) ***08/2025–Present***

Master of Science in Robotics | GPA: 4.07/5.00 | Expected graduation: Nov 28, 2026

- Research focus: autonomous exploration, reinforcement learning, decision optimization, viewpoint planning, and graph-based policy learning.
- Relevant coursework: Topics in Robotics (A+), AI Planning and Decision Making, Cognitive Robotics, Robot Dynamics and Control, and Machine Learning in Robotics.

Shandong University (SDU) ***09/2021–06/2025***

B.Eng. in Optical Information Science and Engineering | GPA: 3.61/5.00 | Rank: 7/43

RESEARCH EXPERIENCE

Preference-Adaptive Multi-Agent Optimization for Multi-Objective Combinatorial Optimization ***04/2026–Present***

Manuscript under review, MARMoT Lab, National University of Singapore

Authors: **Luode Shi***, Conghui Wang, Mingfeng Fan, and Guillaume Adrien Sartoretti*

- Developed PAMAO for multi-agent routing and scheduling; integrated agent-wise preference adaptation (AWPI) and Collaborative Self-Improvement (CSI) into MACSIM, PARCO-WECA, and DCAN variants.
- **MOMTSP:** Across nine generated settings, average per-setting HV gains were about 3.1% vs. MACSIM-WECA and 5.6% vs. PARCO-WECA. On 24 public cases, mean HV gains were 2.9% for PAMAO-M vs. MACSIM-WECA and 6.5% for PAMAO-P vs. PARCO-WECA at 5/8 vehicles.
- **MOHCVRP:** PAMAO-M and PAMAO-P improved mean HV over their matched baselines across nine generated settings; PAMAO-M also showed a modest gain over MACSIM-WECA on 20 public cases.
- **MOMAFJSP:** PAMAO-M and PAMAO-D improved mean HV over MACSIM-WECA and DCAN, respectively, on generated two-objective ID settings. For three-objective DCAN tests, PAMAO-D raised mean normalized HV by 28.9% on ID and 25.4% on OOD scales (six each), using DCAN's official greedy protocol (one rollout per preference). Recorded evaluation time: 0.20–1.21 s/case for PAMAO-D vs. 1.76–20.49 s/case for DCAN (hardware-dependent).

Ariadne: Reinforcement Learning for Single-Robot Autonomous Exploration ***12/2025–05/2026***

Research Project, MARMoT Laboratory, National University of Singapore

Advisor: Asst. Prof. Guillaume Sartoretti | Robotics, Autonomous Exploration, Reinforcement Learning

- Built a single-robot exploration system that selects observation viewpoints to reveal unknown areas in grid maps; the robot moves one A* step, updates its map, and replans at each decision.
- Clustered frontiers and sampled Gaussian viewpoints to create candidate actions; snapped candidates to traversable graph nodes, removed duplicates, ranked them by exploration utility, and capped the set at 25 per decision.
- Implemented a Transformer-pointer policy with twin-critic Soft Actor-Critic. In the expert-supervised setup, a ground-truth Gaussian-viewpoint planner selected frontier-covering viewpoints and ordered a route; an imitation reward aligned policy choices with expert viewpoints and frontier clusters.
- Integrated Ray-parallel rollouts and evaluated the best checkpoint on 100 test maps (up to 600 decisions per run), with a mean travel distance of 517.1 m. HEADER reports 529 ± 67 m on its 100-scenario test set (arXiv:2510.15679).

ADDITIONAL EXPERIENCE

Machine Vision Intern, Shandong Institutes of Industrial Technology ***07/2024–08/2024***

Industrial inspection and machine vision

- Built a production-line inspection workflow for open glass bottles, from camera calibration and image threshold setup through defect filtering and result reporting.
- Used MXVision and image-region size/shape rules to identify empty positions, inverted bottles, and bottles that had not been opened; tuned the checks with production samples.
- Processed inspection results with C# scripts and delivered bottle counts to the client; also assisted with defect-inspection tasks for Hikrobot products.

TECHNICAL SKILLS

- **Reinforcement Learning:** decision-process modeling, Soft Actor-Critic, twin value networks, reward design, expert-route imitation, and policies guided by objective trade-offs.
- **Multi-Agent Optimization:** multi-agent routing and job-shop scheduling; preference-guided neural solvers, agent-role adaptation, collaborative training, and solution-quality evaluation.
- **Robotics and Exploration:** occupancy-map exploration, frontier and viewpoint selection, graph-based planning, candidate-action generation, route coverage, and A*-based path planning.
- **Machine Learning Systems:** Python, PyTorch, Transformer models, Ray, Hydra, TensorBoard, Lightning/RL4CO; experience training and evaluating models on an 8-GPU NVIDIA A100 server, parallel simulation, and unit testing.
- **Development Tools:** AI-assisted coding tools for code reading, experiment scripting, and debugging.
- **Languages:** IELTS 6.5; CET-4 and CET-6.

HONORS AND AWARDS

Optoelectronics Alumni Scholarship, School of Information Science and Engineering, Shandong University ***2025***

Second-Class Academic Scholarship, Shandong University ***2024***

Outstanding Participant, SIOM Summer Camp, University of Chinese Academy of Sciences ***2024***

Second-Class Academic Scholarship, Shandong University ***2023***

Third-Class Academic Scholarship, Shandong University ***2022***